

Fake3DGS: A Benchmark for 3D Manipulation Detection in Neural Rendering

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Motivations

Deepfakes: images or videos manipulated to distort reality or modify the features and actions of filmed people. **Totally synthetic images** obtained from textual prompts that can represent whatever the creator wishes are, thus, **fakes**.



Rendered images from NeRF and Gaussian Splatting models:
fake (i.e., generated) or real?

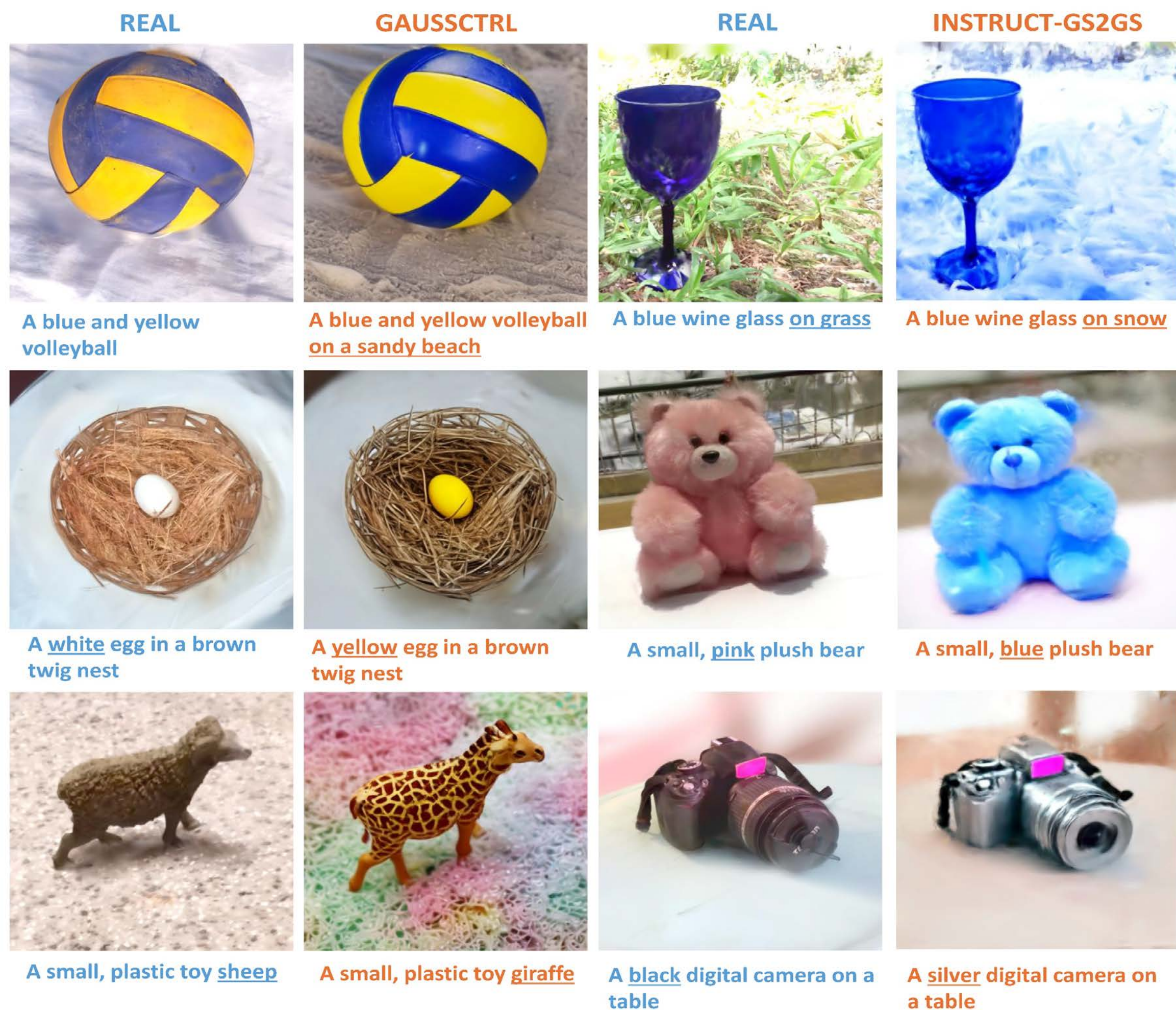
Rendered images from NeRF or GS are AI generated but they belong to a real scene. Fakes can be generated by manipulating the GS scene, leaving unchanged the image after rendering.

Unlike 2D manipulations, edits performed directly in the 3D scene representation can produce an unlimited number of photorealistic, view-consistent images, making traditional artifact-based detection strategies limited in efficacy.

This shift introduces a **new class of deep fakes**, where geometric consistency becomes an advantage for the attacker rather than a vulnerability.

The Fake3DGS Dataset

- Fake3DGS dataset, a large-scale benchmark of original and edited 3D Gaussian Splatting scenes.
- The dataset comprises more than 41k reconstructed scenes, evenly balanced between real and manipulated samples.
- We export all reconstructions as compressed nerfstudio2 checkpoints.
- Compression: post-process with Self-Organizing Gaussian Grids (SOGS) [38] and Gaussian parameter quantization (uniform scalar quantization - 8-bit)
- Shrinking of the dataset footprint from approximately 7TB to about 200 GB.



Scene manipulation

- We started from UCO3D [34], an existing large-scale dataset that already provided a large number of scenes, each with multi-view imagery, accurate camera poses, and a stable 3D scene representation for evaluation.
- The edits cover the three instruction families used in our benchmark background/surface changes (e.g., grass → snow), object appearance changes (e.g., color), and object type/material substitutions (e.g., sheep → giraffe) while keeping the overall scene visually plausible.
- Sample Prompt: *Modify the following sentence by changing the material or the type of the main object, but do not change the color, the background, or the shape.*

References

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Fake detection algorithms

Baselines:

- CoDE (Contrastive Deepfake Embeddings) [2], which learns a deepfake oriented embedding space via contrastive learning while enforcing global/local agreement between full-image and cropped-view representations to capture both global cues and fine-grained artifacts.
- CLIP-based baselines following Ojha et al. [41]: we load pretrained CLIP encoders [43] and train a linear classifier on top of the CLIP image embedding.
- self-supervised visual backbones based on DINOv2 [42], by fine-tuning the corresponding pretrained representations with an identical linear classification head.

Fake3DD (Our solution)

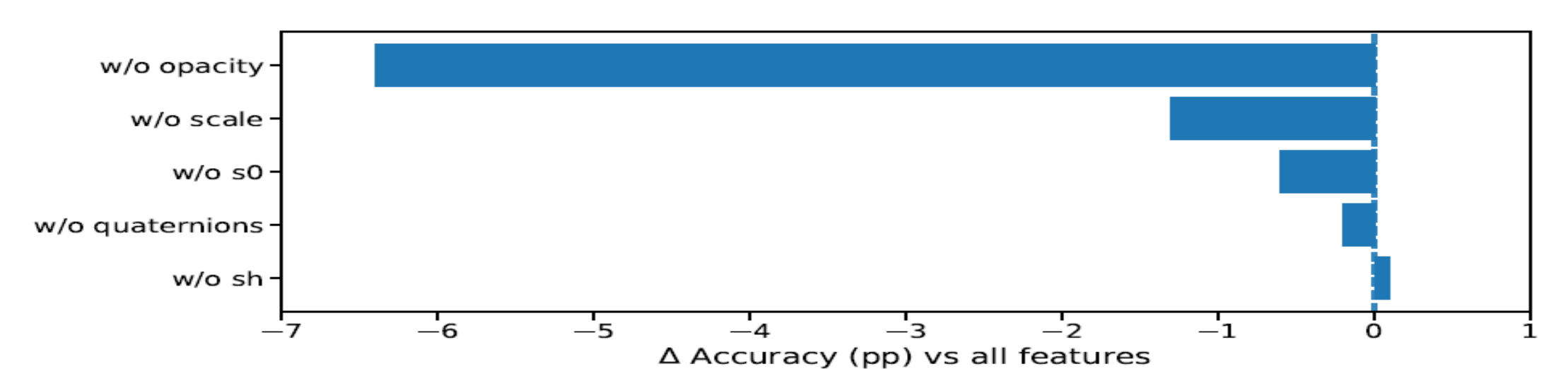
Fake3DD operates directly on the Gaussian splatting representation. Backbone: **PointTransformerV3** model [53], a transformer-based architecture for point clouds that treats a point set as an unordered collection of 3D samples. **Input feature representation:** 3D coordinates of each Gaussian mean point, opacity, scale, quaternions, spherical harmonics sh with the zeroth-order term s0 representing the view-independent color component.

Given that each 3D scene is represented by a variable number of Gaussian splats, these Gaussian-level features are aggregated into a single scene-level representation using a global mean pooling operation. Features belonging to the same scene are averaged based on their batch indices, producing a fixed-dimensional embedding per scene. These embeddings are then passed to a classification head for the final binary prediction (**fake or real classes**).

Experimental results

Backbone	Train		Test		Accuracy (%)		
	GaussCtrl	Instruct-GS2GS	GaussCtrl	Instruct-GS2GS	Overall	Fake	Real
CLIP ViT-B	✓			✓	80.7	70.6	95.9
CLIP ViT-B		✓	✓		74.3	41.5	98.5
CLIP ViT-B	✓	✓	✓	✓	84.0	84.2	93.8
CLIP ViT-L	✓			✓	81.5	67.1	96.6
CLIP ViT-L		✓	✓		75.4	44.5	98.2
CLIP ViT-L	✓	✓	✓	✓	84.0	83.7	94.3
DINOv2-B	✓			✓	76.8	72.7	89.2
DINOv2-B		✓	✓		73.2	45.9	93.3
DINOv2-B	✓	✓	✓	✓	87.8	89.7	85.8
CoDE [2]	✓			✓	80.2	60.5	92.4
CoDE [2]		✓	✓		79.3	54.1	95.8
CoDE [2]	✓	✓	✓	✓	92.2	91.4	92.9
DM [12]	✓			✓	83.8	74.8	95.8
DM [12]		✓	✓		82.4	80.3	96.9
DM [12]	✓	✓	✓	✓	90.4	89.3	97.2
UFD [41]	✓			✓	79.5	69.4	91.9
UFD [41]		✓	✓		78.2	59.5	95.4
UFD [41]	✓	✓	✓	✓	89.9	90.6	89.2
Fake3DD (Ours)		✓	✓		98.2	97.8	98.6
Fake3DD (Ours)	✓			✓	98.6	99.1	98.3
Fake3DD (Ours)	✓	✓	✓	✓	98.9	98.1	99.5

Experimental results under different train/test split protocols. We report overall accuracy together with class-wise accuracy on fake and real samples. For each backbone, we evaluate (i) cross-edit generalization by training on one editing method (GaussCtrl or Instruct-GS2GS) and testing on the other, and (ii) the mixed setting where training and testing use the combined data



Ablation over Gaussian feature groups. We report the change in accuracy (in percentage points) obtained by removing each feature group from the input, relative to the full-feature model

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